
The Limit of a Function

Work these in class. Worked solutions are on the posted notes page — each problem below names the section heading to look for.

2.2 • Problem 1*see: Estimating a Limit from a Table*

Two different questions. **What is $f(1)$?** And **what is this table heading for?** Answer both — then say which of the two the limit is asking.

2.2 • Problem 2*see: Reading Limits off a Graph*

From the graph: $\lim_{x \rightarrow 1} f(x)$, $f(1)$, $\lim_{x \rightarrow 3} f(x)$, $\lim_{x \rightarrow 5} f(x)$, $f(5)$. Say if one does not exist.

2.2 • Problem 3*see: When a Table Lies to You*

Compute $f(x)$ at $x = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$. What would you conclude?

2.2 • Problem 4*see: Approaching from One Side*

Go back to the graph. What are $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$?

2.2 · Problem 5 (worked together)*see: Limits That Grow Without Bound*

Look at $g(x) = \frac{1}{x^2}$ as $x \rightarrow 0$. As x gets small, x^2 gets very small and positive, so $1/x^2$ gets enormous — from both sides.

2.2 · Problem 6*see: Limits That Grow Without Bound*

Now try $h(x) = \frac{1}{x}$ as $x \rightarrow 0$. Check both sides separately. Is the answer the same?

2.2 · Problem 7*see: Vertical Asymptotes*

Find the vertical asymptotes of $f(x) = \frac{x+1}{(x-2)(x+3)}$. Then decide the sign of f just to the right of $x = 2$.